

Jasoliya Parthiv Mehulkumar

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Education

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat

2022 – 2026

B.Tech in Electronics & Communication Engineering · CGPA: 8.21/10

Experience

TruEstate – Software Engineering Intern | *Next.js, NestJS, TypeScript, GCP, Qdrant*

Jan 2026 – Jul 2026

- Designed and deployed cloud-native backend services for property search, data processing, and asynchronous workflows using Cloud Run, Cloud SQL, Pub/Sub, Cloud Tasks, BigQuery, and other GCP services for **alldoors**.
- Launched a high-performance property matching engine using vector embeddings and Qdrant, computing semantic similarity across property listings to deliver intelligent property match scores for buyers.
- Built an AI-powered internal email platform with semantic search, predictive assistance, and generative AI integrations for sales workflows.

TCS R&I – Software Engineering Intern | *Python, AWS, LLMs, Agentic Frameworks*

May 2025 – Jul 2025

- Designed a hybrid ingestion pipeline using Python for high-volume structured feeds (STIX/JSON) and a fine-tuned LLM to extract entities from unstructured vendor reports, processing 100,000+ events/day.
- Automated CIS benchmark checks across 50+ AWS resources using Python Lambda agents, identifying and mitigating 120+ security misconfigurations pre-deployment.
- Evaluated 10,000+ IAM access patterns using custom AI agents, improving IAM policy hygiene across AWS environments.

Stealth – Software Engineering Intern | *Python, CUDA, PyTorch, Diffusion Models, GANs*

Jul 2024 – Aug 2024

- Architected a Stable Diffusion and PyTorch pipeline to generate 500+ 2D game assets per hour, seamlessly integrating colliders and rigid-body parameters into the game engine.
- Optimized the generation workflow by batching denoising steps and improving GPU utilization, reducing latency from 6.0s to 5.1s across a 1,000-asset benchmark while maintaining consistent visual fidelity.
- Enhanced environment pattern generation, increasing designer-approval rate from 72% to 79% through A/B testing.

Projects

Strata – Quant Trading Platform | *Python, C++, Polars, React ([GitHub](#))*

- Developed a high-performance C++ Monte Carlo engine exposed to Python via PyBind11, simulating 10,000+ price paths for VaR and CVaR risk estimation.
- Implemented Hierarchical Risk Parity using unsupervised clustering to construct stable portfolios, reducing out-of-sample variance by 15% compared to traditional optimization across a 5-year backtest.
- Built a statistical arbitrage service using Engle–Granger cointegration to identify mean-reverting pairs from 500+ equities, achieving a simulated Sharpe ratio of 1.85.

Graph Visualizer – Algorithm Analysis Tool | *React, TypeScript, Tailwind CSS ([Live](#))*

- Built a graph editor supporting multiple input formats (edge lists, adjacency lists, parent–child arrays).
- Implemented graph analysis features including components/SCC, MST, bridges & articulation points, and tree/bipartite visualizations and garnered over 60 users.
- Optimized force-layout rendering to smoothly visualize graphs with ~1000 nodes and added interactive modes (lock, color, multi-testcases).

Technical Skills

Languages: C++, Python, Java, TypeScript, JavaScript, HTML5, CSS3.

Backend: REST APIs, FastAPI, Node.js, Express, Spring Boot, NestJS.

Cloud & DevOps: GCP, AWS, Docker, Git, GitHub, GitLab, Postman.

Databases: MySQL, MongoDB, Firebase, Qdrant.

Frontend: React, Next.js.

AI / Tools: LLMs, Agentic Frameworks, Stable Diffusion, Polars.

Security: Ghidra, JADX.

Achievements

- LeetCode:** Max Rating 2409 (**Guardian**) – 600+ Problems Solved. [[Profile](#)]
- Ranked **#130/22,000+** in **Meta Hacker Cup 2025 R3** and **#581** in 2024 R1. [[Certificate](#)]
- ICPC Regionalist** (Amritapuri Regional Round) – 2024 and 2025. [[Certificate](#)]
- Hackathon Finalist** – Precision Healthcare Challenge 2025 (GE Healthcare).
- Amazon ML Summer School 2024** – Selected among **3,000** participants.